

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – NOVEMBER, 2014.



Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself.
5. Maximum time allowed is THREE hours.
6. All questions carry equal marks.

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1. A hare, pursued by a grey-hound, is 50 of her own leaps ahead of him. While the hare takes 4 leaps the grey-hound takes 3 leaps. In one leap the hare goes $1\frac{3}{4}$ metre and the grey-hound $2\frac{3}{4}$ metre. In how many leaps will the grey-hound overtake the hare?
 2. If $\sqrt{a-x} + \sqrt{b-x} + \sqrt{c-x} = 0$, show that $(a+b+c+3x)(a+b+c-x) = 4(ab+bc+ca)$.
 3. Some amount of work has to be completed. *Anand*, *Bilal* and *Charles* offered to do the job. *Anand* would alone take '*a*' times as many days as *Bilal* and *Charles* working together: *Bilal* would alone take *b* times as many days as *Anand* and *Charles* together. *Charles* would alone take *c* times as many days as *Anand* and *Bilal* together. Show that $\frac{a}{a+1} + \frac{b}{b+1} + \frac{c}{c+1} = 2$.
 4. Let *P* be any point on the diagonal *BD* of a rectangle *ABCD*. *F* is the foot of the perpendicular from *P* to *BC*. *H* is a point on the side *BC* such that *FB* = *FH*. *PC* cuts *AH* in *Q*. Show that Area of $\triangle APQ$ = Area of $\triangle CHQ$.
 5. A number of three digits in scale 7 when expressed in scale 9 has its digits reversed in order. Find the number in scale 7 and scale 10.
 6. a) Two regular polygons have the number of their sides as 3:2 and the interior angles as 10:9. Find the number of sides of the polygons.
b) Find two natural numbers such that their difference, sum and the product is to one another as 1,7 and 24.
 7. Find the value of *a*, *b*, *c* which will make each of the expressions $x^4 + ax^3 + bx^2 + cx + 1$ and $x^4 + 2ax^3 + 2bx^2 + 2cx + 1$ a perfect square.
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